

Probit

experimental, 5-member ensemble, evaluated on 2016-2026, measured against the market

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Model identity

Every figure below is read from a declared artifact. This report fits nothing, loads no booster, recomputes no prediction and calls no betting optimiser.

MODEL Probit experimental	EVALUATION YEARS 11 2016-2026	ENSEMBLE MEMBERS 5 seeds 7, 17, 42, 123, 256	BOOSTING ROUNDS 100
FEATURE SET production 28 declared inputs			

Probit is experimental.

It is not the production model. Nothing here is a promotion decision, and the promotion gates it can and cannot satisfy are in the technical appendix.

HOW EACH EVALUATION YEAR WAS FITTED

These are walk-forward figures rather than one model's: the model is refitted for each evaluation year on the seasons before it, so the result concatenates 11 separately fitted models and no evaluation year's outcomes appear in its own fit. This run did not record the year its training window starts, so this report does not name one.

Deltas in this report use `model_minus_market_log_loss: race_multinomial_log_loss`, where **negative** is better. That is stated because three delta fields existed in this repository and they did not all point the same way.

Provenance -- the source commit, the config and data digests, the environment and every input this report read by content digest -- is in the technical appendix and in `run_lock.json` beside this file. Canonical publication fails without it.

Executive summary

MEAN DELTA VS MARKET +0.002018 negative is better	RACES 7,933	YEARS BETTER 4/11	PAIRED DELTA -0.012557 against softmax
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Probit is worse than the market by 0.002018 `race_multinomial_log_loss` on 7,933 races. The direction holds in 4 of 11 evaluated years. It is behind the market: on the common yardstick this model loses to the prices it was measured against. There is also a paired diagnostic against softmax, which holds everything but one change fixed: on that comparison the delta is -0.012557. That is the opposite sign to the market delta above. Both are true: this model beats the reference it holds fixed and still loses to the market, and the market is the yardstick every report here shares. Every seed agrees on the sign of that paired delta.

Evaluation population and market reference

THE MARKET REFERENCE

Every metric below is reported beside the market's own implied probabilities on the same races, normalised within each race, and the delta is this model's race log loss minus the market's on the same races. One yardstick across every published report, so their headline numbers mean the same thing and can be read against one another. The market is a measurement, not a model this one is defined against.

THE EVALUATION POPULATION

Every figure in this report covers the standardized evaluation window, which begins in 2016. One window for every published report, so their headline numbers are computed on the identical race keys and can be compared -- which is checked by digest before anything is published, not asserted in a footnote. 2015 is earlier than the window and is excluded from every evaluation figure below, including the bankroll paths. The feature-effect grid is the one exception, and says so where it appears: a learned response direction is a property of each year's fit rather than of the evaluation population.

TEST RACES
7,933

YEARS
11
2016-2026

SEEDS
5

Some years are warm-up rather than evaluation

2015 had no earlier year to fit a utility scale on and took the predeclared theoretical value (insufficient_history). The utility is divided by that scale, so those years amplify corrections where a fitted scale halves them, and their stakes and claimed edges are several times any fitted year's. They are excluded from every figure in this report and named here rather than labelled and left in.

Overall probability performance

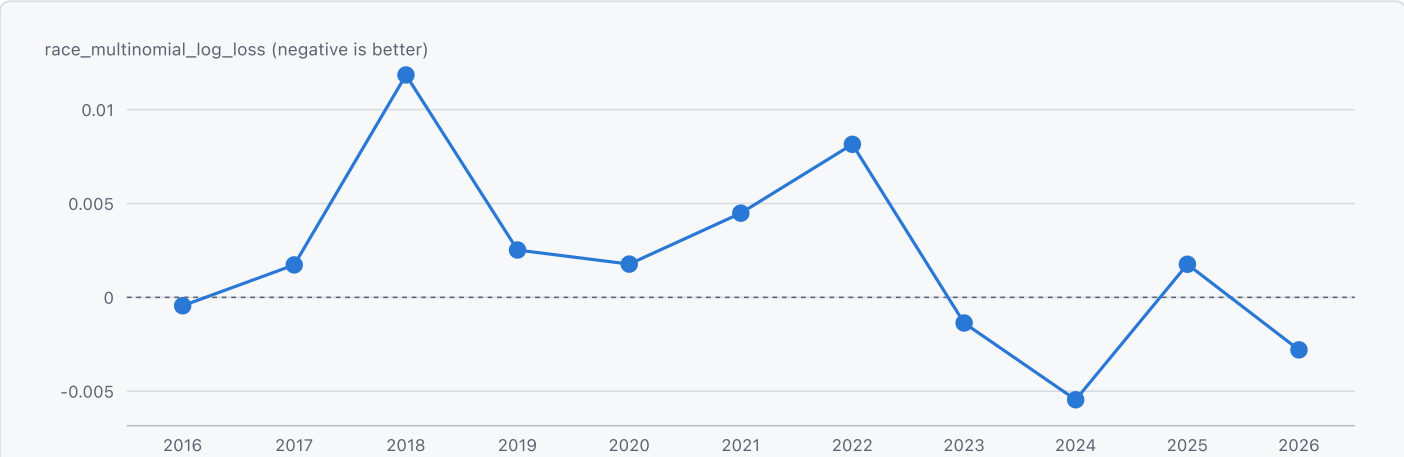
races	model log loss	market log loss	market delta	better year share
7933	2.030823	2.028805	0.002018	0.363636

One row, pooled over every evaluated race, and measured against the market: market_delta is the model's race log loss minus the market's on the same races, so negative means the model is better. The per-race distribution is what the year-by-year section shows; a pooled mean can be carried by a handful of races and this figure alone cannot tell you whether it was.

Year-by-year probability performance

test year	races	model log loss	market log loss	market delta	honest
2016	744	2.098827	2.099277	-0.000450	yes
2017	772	2.023286	2.021559	0.001727	yes
2018	756	2.073354	2.061504	0.011849	yes
2019	759	1.935353	1.932827	0.002526	yes
2020	799	2.012171	2.010400	0.001771	yes
2021	783	2.011146	2.006659	0.004487	yes

test year	races	model log loss	market log loss	market delta	honest
2022	783	2.059349	2.051195	0.008154	yes
2023	796	2.022700	2.024066	-0.001366	yes
2024	792	2.020289	2.025744	-0.005455	yes
2025	748	2.059159	2.057397	0.001763	yes
2026	201	2.023421	2.026224	-0.002803	yes



Mean delta by year Mean delta against the market for each evaluated year. Zero is the market.

▼ Table view

Category	race_multinomial_log_loss (negative is better)
2016	-0.00045
2017	0.001727
2018	0.011849
2019	0.002526
2020	0.001771
2021	0.004487
2022	0.008154
2023	-0.001366
2024	-0.005455
2025	0.001763
2026	-0.002803

Ahead of the market in 4 of 11 years.

Calibration and sharpness

This architecture reports calibration through a paired comparison rather than binned reliability: this model and its reference are each given a temperature on the same races, so the calibrated delta is the like-for-like figure. The reference is named in the paired diagnostic in the technical appendix, together with what the comparison holds fixed. Fallback-scale years are excluded here, as they are from the headline.

test year	this model calibrated loss	reference calibrated loss	calibrated delta
2016	2.096131	2.103251	-0.007120
2017	2.024009	2.031793	-0.007785
2018	2.073354	2.079390	-0.006036
2019	1.938096	1.945433	-0.007337
2020	2.012574	2.024462	-0.011888

test year	this model calibrated loss	reference calibrated loss	calibrated delta
2021	2.011085	2.017480	-0.006395
2022	2.059959	2.065483	-0.005524
2023	2.023223	2.025731	-0.002508
2024	2.021114	2.026537	-0.005423
2025	2.060147	2.066830	-0.006683
2026	2.024664	2.035721	-0.011056

If the calibrated delta is materially smaller than the uncalibrated one, part of this model's advantage was a temperature the reference could also have had.

Top-choice movement

Not applicable to this model. Top-choice movement needs the market's own first pick beside the model's, and this run's artifacts do not carry it. The paired diagnostic in the appendix holds the mean utilities fixed, which is a different and narrower question.

Ensemble stability

Not applicable to this model. There is no ensemble to be stable. This architecture is one choice link applied to one set of mean utilities, not a bag of independently fitted members, so seed variation here is integration noise rather than model disagreement -- and that is reported in the appendix instead.

Feature and correction effects

Not applicable to this model. There are no per-feature effects belonging to this model to attribute. A choice link changes how utilities become probabilities; it has no features of its own. The mean utilities are another run's fit, so attributing their feature effects here would credit this model with that run's work -- and for the market-anchored architecture they are the Softmax Offset margins exactly, whose per-feature effects, response directions and group ablation are in that report. What is specific to this model is the link and its utility scale, and both are in the technical appendix.

Betting and bankroll analysis

Retrospective odds

Every stake, return and bankroll figure in this report is computed against historical result-page odds and official dividends. Those are not an executable pre-race price: they are the prices after the pool closed, and a bet placed at them could not have been placed. Nothing here establishes profitability. A profitability claim requires a locked model shadow-tested against timestamped pre-race odds snapshots.

Do not select a threshold from these returns.

Every year through 2026 has been used as development data in this repository: features were searched, architectures compared and thresholds inspected against these same seasons. The chronological split is real, so a test year is genuinely after its training rows, but no test year is unseen. Treat the forward figures as probability progress rather than as a locked out-of-sample result, and do not select a Kelly level or an edge threshold from this reused history. The confirmation step that this run cannot substitute for is a prospective shadow test against archived pre-race price snapshots.

SETTLEMENT BY STAKING RULE

Role	What it is for
Primary	what an operator would run: sized for the rebate, and credited it.
Sizing reference	the same settlement, sized blind. It isolates sizing for the cliff from receiving the rebate -- this row and the primary settle identically and differ only in what the solver knew.
Settlement ablation	blind sizing with the rebate not credited. It isolates the rebate's own contribution to the return.
Benchmark	a fixed flat stake above an edge floor. Not a Kelly fraction, so it belongs to no preset panel below.

The table below is ordered by that role rather than alphabetically, so the first Kelly rows are the ones an operator would run.

strategy	n bets	turnover	net return	roi	total log growth	max drawdown	final bankroll	underwater days
rebate_aware_kelly@0.01	7715	18,207,680	-2,607,460	-0.143207	-0.191536	0.188440	8256902.450000	4251
rebate_aware_kelly@0.02	7815	35,887,720	-5,181,461	-0.144380	-0.403814	0.352974	6677686.250000	4280
rebate_aware_kelly@0.05	8054	76,249,900	-10236084.500000	-0.134244	-0.935557	0.636599	3923673.600000	4402
rebate_aware_kelly@0.1	7937	105,491,460	-14313770.500000	-0.135687	-2.055409	0.889349	1280404.400000	4352
kelly_settled_rebate@0.01	7369	16,228,700	-2,254,076	-0.138894	-0.162460	0.161965	8,500,497	4254
kelly_settled_rebate@0.02	7405	30,525,200	-4,236,205	-0.138777	-0.323747	0.296708	7,234,332	4275
kelly_settled_rebate@0.05	7418	63,771,520	-8807677.500000	-0.138113	-0.823451	0.590323	4389142.500000	4289
kelly_settled_rebate@0.1	7414	96,951,250	-13,224,599	-0.136405	-1.728009	0.844323	1,776,377	4286
kelly_no_rebate@0.01	7363	15,537,960	-2145786.500000	-0.138100	-0.241535	0.221390	7854213.500000	4256
kelly_no_rebate@0.02	7399	28,029,580	-3,860,391	-0.137726	-0.487824	0.396485	6,139,609	4275
kelly_no_rebate@0.05	7407	52,710,070	-7,124,518	-0.135164	-1.246365	0.723988	2,875,482	4282
kelly_no_rebate@0.1	7377	71,427,330	-9,245,495	-0.129439	-2.584278	0.930033	754,505	4262
flat_stake@edge>=0.02	6510	65,100,000	-25,665,000	-0.394240	-36.841361	2.531206	-15,665,000	3833

Model view bagged_probit. Seed -1. These rules see the same races and the same odds. They do not necessarily choose the same runners or the same races: rebate-aware sizing admits runners a blind solve would reject, because staking a mildly negative-edge runner raises the loss on every other outcome and can push a race over the rebate threshold. Compare the bet counts as well as the returns.

What this settlement can decide

Secondary diagnostic. This suite selects on paired race log loss; betting returns cannot select a model and are not evidence for or against a probit candidate.

EXPECTED AGAINST REALISED, BY ESTIMATED EDGE

FLAT STAKE

bucket	mean estimated edge	expected return	realised at edge>=0.02	bets at edge>=0.02
0.020000 to 0.050000	0.034377	0.034377	-0.1738	951
0.050000 to 0.100000	0.072830	0.072830	0.0143	1002
0.100000 to 0.200000	0.144020	0.144020	-0.3538	1230
0.200000 to unbounded	0.578110	0.578110	-0.5952	3327

KELLY, NO REBATE

bucket	mean estimated edge	expected return	realised at 0.01	realised at 0.02	realised at 0.05	realised at 0.1	bets at 0.01	bets at 0.02	bets at 0.05	bets at 0.1
0 to 0.020000	0.010508	0.010508	-0.2136	-0.2361	-0.2906	-0.4206	853	889	897	867
0.020000 to 0.050000	0.034377	0.034377	-0.0045	-0.0227	-0.0838	-0.2033	951	951	951	951
0.050000 to 0.100000	0.072830	0.072830	-0.2092	-0.2113	-0.2121	-0.1886	1002	1002	1002	1002
0.100000 to 0.200000	0.144020	0.144020	-0.0966	-0.0846	-0.0509	-0.0084	1230	1230	1230	1230
0.200000 to unbounded	0.578110	0.578110	-0.1469	-0.1472	-0.1465	-0.1431	3327	3327	3327	3327

KELLY, SETTLED REBATE

bucket	mean estimated edge	expected return	realised at 0.01	realised at 0.02	realised at 0.05	realised at 0.1	bets at 0.01	bets at 0.02	bets at 0.05	bets at 0.1
0 to 0.020000	0.010483	0.010483	-0.2098	-0.2203	-0.2503	-0.3224	859	895	908	904
0.020000 to 0.050000	0.034377	0.034377	0.0008	-0.0120	-0.0512	-0.1307	951	951	951	951
0.050000 to 0.100000	0.072830	0.072830	-0.2098	-0.2105	-0.2141	-0.2102	1002	1002	1002	1002
0.100000 to 0.200000	0.144020	0.144020	-0.1008	-0.0930	-0.0693	-0.0338	1230	1230	1230	1230
0.200000 to unbounded	0.578110	0.578110	-0.1475	-0.1479	-0.1489	-0.1495	3327	3327	3327	3327

REBATE-AWARE KELLY

bucket	mean estimated edge	expected return	realised at 0.01	realised at 0.02	realised at 0.05	realised at 0.1	bets at 0.01	bets at 0.02	bets at 0.05	bets at 0.1
unbounded to 0	-0.031497	-0.031497	-0.3107	-0.3763	-0.1766	-0.2306	351	412	643	531
0 to 0.020000	0.010503	0.010503	-0.2290	-0.1475	-0.1207	-0.1626	854	893	901	896
0.020000 to 0.050000	0.034377	0.034377	0.0163	-0.0149	-0.0791	-0.1452	951	951	951	951
0.050000 to 0.100000	0.072830	0.072830	-0.2118	-0.1868	-0.1819	-0.1980	1002	1002	1002	1002
0.100000 to 0.200000	0.144020	0.144020	-0.0853	-0.0891	-0.0686	-0.0368	1230	1230	1230	1230

bucket	mean estimated edge	expected return	realised at 0.01	realised at 0.02	realised at 0.05	realised at 0.1	bets at 0.01	bets at 0.02	bets at 0.05	bets at 0.1
0.200000 to unbounded	0.578110	0.578110	-0.1488	-0.1504	-0.1483	-0.1447	3327	3327	3327	3327

The check that matters most in this section: if the realised return in each edge bucket does not track the expected one, the edge estimate is wrong and every stake sized from it is wrong with it. The bucket counts are small enough that one winner moves a bucket, so read the pattern across buckets rather than any single number.

A negative-edge bucket appears only under rebate-aware sizing, and only at the larger fractions. That is the rule working as defined rather than a filter failing: the rebate is a cliff, so a runner whose own edge is negative can pay for itself by pushing the race's turnover over the threshold, and the stakes only get large enough for that at the higher fractions.

KELLY PRESETS

OPENING BANKROLL HK\$10,000,000	PRESETS SETTLED 1% Kelly, 2% Kelly, 5% Kelly, 10% Kelly	PATH MODES annual reset, continuous	SEEDS 5 7, 17, 42, 123, 256
TRAINED ON 11 separate fits, one per evaluation year the training start year is not recorded in this run's metadata	BOOSTING ROUNDS 100	YEARS STAKED 2016-2026 2015 excluded	

These are walk-forward figures rather than one model's: the model is refitted for each evaluation year on the seasons before it, so the result concatenates 11 separately fitted models and no evaluation year's outcomes appear in its own fit. This run did not record the year its training window starts, so this report does not name one. Each path is rebased on its own opening balance, so zero means break-even for every line from the same point. Where a year is excluded, a continuous path opens at the balance the bank held when the kept years begin and an annual-reset path opens at the declared bankroll, so the two show different opening figures for the same rule. Rebate policy: 10% of the race net loss, once that loss reaches HK\$10,000, as a cliff. Flat staking is omitted from these panels. Its stake is a fixed amount per bet and this settlement never halts on ruin, so its path can pass below zero -- a measure of how badly a rule loses rather than a balance an account could hold. It is also not a Kelly fraction, so it belongs to no preset. Its settlement is in the strategy table above. 2015 is excluded from every path below. These test years have no earlier year to fit the utility scale on and fall back to the predeclared theoretical value. The utility is divided by the scale, so the fallback amplifies corrections where a fitted scale halves them: their stakes and claimed edges are several times any fitted year's and are not comparable with them. Exclude these years from any return figure -- filter on `is_honest_scale` in the per-run CSV.

1% KELLY

OPENING BANKROLL

HK\$8,897,664

PRIMARY FINAL RETURN

-7.2%

Rebate-aware Kelly,
continuous

BETS

6,283

settled bets

REBATE RECEIVED

HK\$62,394

by the primary rule

Rebate-aware Kelly Kelly, settled rebate Kelly, no rebate



Cumulative bankroll return at 1% Kelly Stepped at each settlement: a bankroll moves only when a bet settles, so the path is flat between settlements and a smooth line would draw growth on days nothing was staked. Dashed lines are the annual-reset mode. Retrospective odds.

Table view

Series	Settlements	Final return	Note
Rebate-aware Kelly, annual reset	3,525	-0.122%	6,283 bets
Rebate-aware Kelly, continuous	3,525	-7.2%	6,283 bets
Kelly, settled rebate, annual reset	3,528	-0.125%	6,127 bets
Kelly, settled rebate, continuous	3,528	-6.3%	6,127 bets
Kelly, no rebate, annual reset	3,525	-0.116%	6,121 bets
Kelly, no rebate, continuous	3,525	-6.3%	6,121 bets

BETS PER YEAR

Year	Rebate-aware Kelly	Kelly, settled rebate	Kelly, no rebate
2016	773	751	751
2017	911	883	881
2018	811	789	789
2019	1,068	1,038	1,038
2020	980	965	964
2021	583	571	570
2022	285	279	279
2023	236	230	230
2024	308	299	298
2025	267	261	261

Year	Rebate-aware Kelly	Kelly, settled rebate	Kelly, no rebate
2026	61	61	60

2026 is a season still in progress, italicised: a shorter year, not a decline in activity. Rebate-aware Kelly: 773 bets in 2016 against 267 in 2025. Kelly, settled rebate: 751 bets in 2016 against 261 in 2025. Kelly, no rebate: 751 bets in 2016 against 261 in 2025. Where the path is flat and these counts are low, the model stopped finding bets rather than losing an edge it had.

Rebate received: Rebate-aware Kelly, annual reset HK\$62,394; Rebate-aware Kelly, continuous HK\$62,394; Kelly, settled rebate, annual reset HK\$4,864; Kelly, settled rebate, continuous HK\$4,864. A rule that reaches the threshold gets money back on a losing race, which makes some races worth betting that otherwise are not -- which is why its bet count can differ from the others'. No staking rule at this fraction converts the model's edge into money on these odds. Both path modes are shown: continuous carries the bankroll across years, while annual-reset restarts it each year, so the reset mode measures per-year skill without compounding.

2% KELLY

OPENING BANKROLL
HK\$7,857,632

PRIMARY FINAL RETURN
-15.0%
Rebate-aware Kelly,
continuous

BETS
6,383
settled bets

REBATE RECEIVED
HK\$340,791
by the primary rule

Rebate-aware Kelly Kelly, settled rebate Kelly, no rebate



Cumulative bankroll return at 2% Kelly Stepped at each settlement: a bankroll moves only when a bet settles, so the path is flat between settlements and a smooth line would draw growth on days nothing was staked. Dashed lines are the annual-reset mode. Retrospective odds.

Table view

Series	Settlements	Final return	Note
Rebate-aware Kelly, annual reset	3,553	-0.464%	6,383 bets
Rebate-aware Kelly, continuous	3,553	-15.0%	6,383 bets
Kelly, settled rebate, annual reset	3,549	-0.214%	6,163 bets
Kelly, settled rebate, continuous	3,549	-12.1%	6,163 bets
Kelly, no rebate, annual reset	3,544	-0.182%	6,157 bets
Kelly, no rebate, continuous	3,544	-12.4%	6,157 bets

BETS PER YEAR

Year	Rebate-aware Kelly	Kelly, settled rebate	Kelly, no rebate
2016	790	754	754
2017	921	886	886
2018	832	795	794
2019	1,071	1,041	1,040
2020	993	968	968
2021	591	575	574
2022	295	285	285
2023	237	231	231
2024	316	305	302
2025	275	262	262
2026	62	61	61

2026 is a season still in progress, italicised: a shorter year, not a decline in activity. Rebate-aware Kelly: 790 bets in 2016 against 275 in 2025. Kelly, settled rebate: 754 bets in 2016 against 262 in 2025. Kelly, no rebate: 754 bets in 2016 against 262 in 2025. Where the path is flat and these counts are low, the model stopped finding bets rather than losing an edge it had.

Rebate received: Rebate-aware Kelly, annual reset HK\$340,791; Rebate-aware Kelly, continuous HK\$340,791; Kelly, settled rebate, annual reset HK\$28,170; Kelly, settled rebate, continuous HK\$28,170. A rule that reaches the threshold gets money back on a losing race, which makes some races worth betting that otherwise are not -- which is why its bet count can differ from the others'. No staking rule at this fraction converts the model's edge into money on these odds. Both path modes are shown: continuous carries the bankroll across years, while annual-reset restarts it each year, so the reset mode measures per-year skill without compounding.

5% KELLY

OPENING BANKROLL

HK\$5,347,723

PRIMARY FINAL RETURN

-26.6%

Rebate-aware Kelly,
continuous

BETS

6,622

settled bets

REBATE RECEIVED

HK\$965,345

by the primary rule

Rebate-aware Kelly Kelly, settled rebate Kelly, no rebate



Cumulative bankroll return at 5% Kelly Stepped at each settlement: a bankroll moves only when a bet settles, so the path is flat between settlements and a smooth line would draw growth on days nothing was staked. Dashed lines are the annual-reset mode. Retrospective odds.

▼ Table view

Series	Settlements	Final return	Note
Rebate-aware Kelly, annual reset	3,675	-0.524%	6,622 bets
Rebate-aware Kelly, continuous	3,675	-26.6%	6,622 bets
Kelly, settled rebate, annual reset	3,562	-0.325%	6,176 bets
Kelly, settled rebate, continuous	3,562	-27.0%	6,176 bets
Kelly, no rebate, annual reset	3,551	-0.213%	6,165 bets
Kelly, no rebate, continuous	3,551	-28.4%	6,165 bets

BETS PER YEAR

Year	Rebate-aware Kelly	Kelly, settled rebate	Kelly, no rebate
2016	849	754	754
2017	972	888	886
2018	862	797	796
2019	1,087	1,043	1,041
2020	1,006	968	968
2021	612	579	576
2022	311	287	285
2023	252	231	231
2024	324	305	305
2025	280	262	262
2026	67	62	61

2026 is a season still in progress, italicised: a shorter year, not a decline in activity. Rebate-aware Kelly: 849 bets in 2016 against 280 in 2025. Kelly, settled rebate: 754 bets in 2016 against 262 in 2025. Kelly, no rebate: 754 bets in 2016 against 262 in 2025. Where the path is flat and these counts are low, the model stopped finding bets rather than losing an edge it had.

Rebate received: Rebate-aware Kelly, annual reset HK\$965,345; Rebate-aware Kelly, continuous HK\$965,345; Kelly, settled rebate, annual reset HK\$96,858; Kelly, settled rebate, continuous HK\$96,858. A rule that reaches the threshold gets money back on a losing race, which makes some races worth betting that otherwise are not -- which is why its bet count can differ from the others'. No staking rule at this fraction converts the model's edge into money on these odds.

Both path modes are shown: continuous carries the bankroll across years, while annual-reset restarts it each year, so the reset mode measures per-year skill without compounding.

10% KELLY

OPENING BANKROLL

HK\$2,620,569

PRIMARY FINAL RETURN

-51.1%

Rebate-aware Kelly,
continuous

BETS

6,505

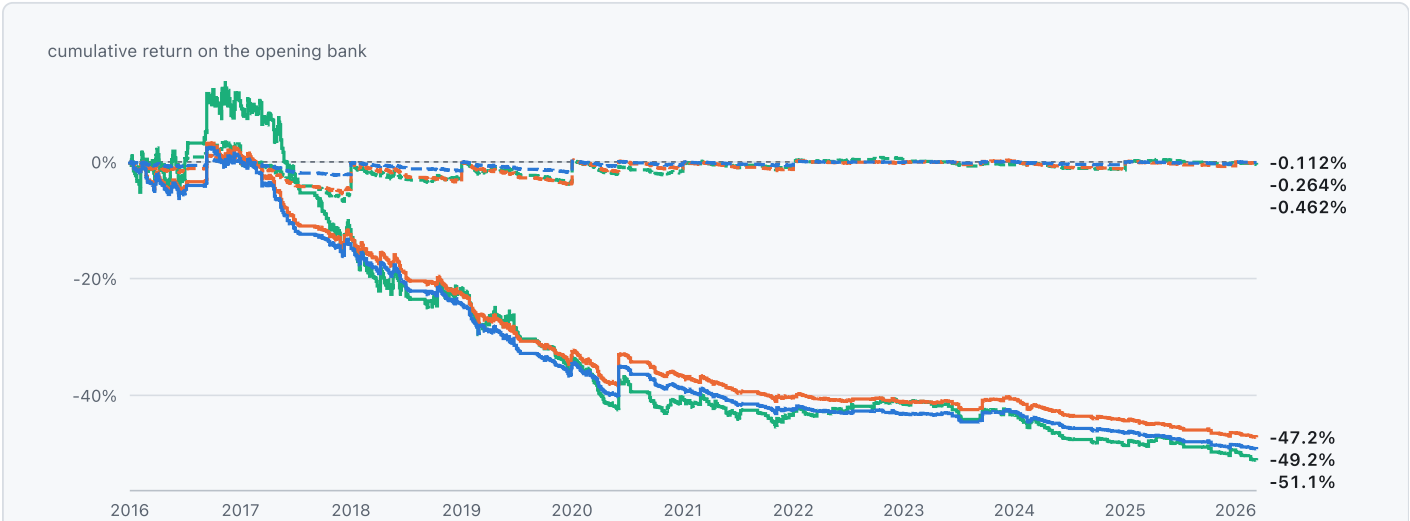
settled bets

REBATE RECEIVED

HK\$684,931

by the primary rule

Rebate-aware Kelly Kelly, settled rebate Kelly, no rebate



Cumulative bankroll return at 10% Kelly Stepped at each settlement: a bankroll moves only when a bet settles, so the path is flat between settlements and a smooth line would draw growth on days nothing was staked. Dashed lines are the annual-reset mode. Retrospective odds.

Table view

Series	Settlements	Final return	Note
Rebate-aware Kelly, annual reset	3,625	-0.462%	6,505 bets
Rebate-aware Kelly, continuous	3,625	-51.1%	6,505 bets
Kelly, settled rebate, annual reset	3,558	-0.264%	6,172 bets
Kelly, settled rebate, continuous	3,558	-47.2%	6,172 bets
Kelly, no rebate, annual reset	3,531	-0.112%	6,135 bets
Kelly, no rebate, continuous	3,531	-49.2%	6,135 bets

BETS PER YEAR

Year	Rebate-aware Kelly	Kelly, settled rebate	Kelly, no rebate
2016	849	754	754
2017	962	888	885
2018	846	797	793
2019	1,076	1,041	1,039
2020	994	968	965
2021	595	578	571
2022	295	287	279

Year	Rebate-aware Kelly	Kelly, settled rebate	Kelly, no rebate
2023	238	231	230
2024	317	305	298
2025	271	262	261
2026	62	61	60

2026 is a season still in progress, italicised: a shorter year, not a decline in activity. Rebate-aware Kelly: 849 bets in 2016 against 271 in 2025. Kelly, settled rebate: 754 bets in 2016 against 262 in 2025. Kelly, no rebate: 754 bets in 2016 against 261 in 2025. Where the path is flat and these counts are low, the model stopped finding bets rather than losing an edge it had.

Rebate received: Rebate-aware Kelly, annual reset HK\$684,931; Rebate-aware Kelly, continuous HK\$684,931; Kelly, settled rebate, annual reset HK\$97,210; Kelly, settled rebate, continuous HK\$97,210. A rule that reaches the threshold gets money back on a losing race, which makes some races worth betting that otherwise are not -- which is why its bet count can differ from the others'. No staking rule at this fraction converts the model's edge into money on these odds. Both path modes are shown: continuous carries the bankroll across years, while annual-reset restarts it each year, so the reset mode measures per-year skill without compounding.

PRESET SENSITIVITY

Preset	Rule and mode	Opening	Final	Return	Bets	Rebate
1% Kelly	Kelly, no rebate, annual reset	HK\$10,000,000	HK\$9,988,420	-0.116%	6,121	HK\$0
1% Kelly	Kelly, no rebate, continuous	HK\$8,385,931	HK\$7,854,214	-6.3%	6,121	HK\$0
1% Kelly	Kelly, settled rebate, annual reset	HK\$10,000,000	HK\$9,987,516	-0.125%	6,127	HK\$4,864
1% Kelly	Kelly, settled rebate, continuous	HK\$9,076,780	HK\$8,500,497	-6.3%	6,127	HK\$4,864
1% Kelly	Rebate-aware Kelly, annual reset	HK\$10,000,000	HK\$9,987,824	-0.122%	6,283	HK\$62,394
1% Kelly	Rebate-aware Kelly, continuous	HK\$8,897,664	HK\$8,256,902	-7.2%	6,283	HK\$62,394
2% Kelly	Kelly, no rebate, annual reset	HK\$10,000,000	HK\$9,981,834	-0.182%	6,157	HK\$0
2% Kelly	Kelly, no rebate, continuous	HK\$7,009,287	HK\$6,139,609	-12.4%	6,157	HK\$0
2% Kelly	Kelly, settled rebate, annual reset	HK\$10,000,000	HK\$9,978,642	-0.214%	6,163	HK\$28,170
2% Kelly	Kelly, settled rebate, continuous	HK\$8,231,586	HK\$7,234,332	-12.1%	6,163	HK\$28,170
2% Kelly	Rebate-aware Kelly, annual reset	HK\$10,000,000	HK\$9,953,593	-0.464%	6,383	HK\$340,791
2% Kelly	Rebate-aware Kelly, continuous	HK\$7,857,632	HK\$6,677,686	-15.0%	6,383	HK\$340,791
5% Kelly	Kelly, no rebate, annual reset	HK\$10,000,000	HK\$9,978,690	-0.213%	6,165	HK\$0
5% Kelly	Kelly, no rebate, continuous	HK\$4,013,972	HK\$2,875,482	-28.4%	6,165	HK\$0
5% Kelly	Kelly, settled rebate, annual reset	HK\$10,000,000	HK\$9,967,520	-0.325%	6,176	HK\$96,858
5% Kelly	Kelly, settled rebate, continuous	HK\$6,015,459	HK\$4,389,142	-27.0%	6,176	HK\$96,858
5% Kelly	Rebate-aware Kelly, annual reset	HK\$10,000,000	HK\$9,947,649	-0.524%	6,622	HK\$965,345
5% Kelly	Rebate-aware Kelly, continuous	HK\$5,347,723	HK\$3,923,674	-26.6%	6,622	HK\$965,345
10% Kelly	Kelly, no rebate, annual reset	HK\$10,000,000	HK\$9,988,758	-0.112%	6,135	HK\$0
10% Kelly	Kelly, no rebate, continuous	HK\$1,485,870	HK\$754,505	-49.2%	6,135	HK\$0
10% Kelly	Kelly, settled rebate, annual reset	HK\$10,000,000	HK\$9,973,568	-0.264%	6,172	HK\$97,210
10% Kelly	Kelly, settled rebate, continuous	HK\$3,364,848	HK\$1,776,377	-47.2%	6,172	HK\$97,210
10% Kelly	Rebate-aware Kelly, annual reset	HK\$10,000,000	HK\$9,953,850	-0.462%	6,505	HK\$684,931
10% Kelly	Rebate-aware Kelly, continuous	HK\$2,620,569	HK\$1,280,404	-51.1%	6,505	HK\$684,931

The spread across presets is the point: a result that appears at only one Kelly fraction is a result about that fraction rather than about the model. A larger fraction stakes more of the bank on the same edge estimate, so it amplifies both the edge and the error in it.

Simulation and uncertainty

MEAN DELTA
-0.012063

INTERVAL
-0.0140 to
-0.0101
80% level

EXCLUDES ZERO
FAVOURABLY
yes

RACE DAYS
891
39,665 races

REPLICATES
2,000
seed 20260824

A day-block bootstrap: race days are resampled within each test year, so the interval accounts for clustering within a meeting, and the interval lies entirely below zero. It conditions on the fitted model and does *not* include refit uncertainty.

What a simulation cannot tell you

The model-assumed simulation draws winners from the model's own probabilities. If those probabilities are wrong the simulation is confidently wrong in the same direction, so it bounds sampling risk and says nothing about model risk.

Technical appendix and provenance

THE UTILITY SCALE

FITTED MINIMUM
1.3893

FITTED MAXIMUM
1.4534

THEORY
0.7797
the value theory predicts

The fitted scale sits above the value theory predicts, in every year: 1.389 to 1.453 against 0.780 (the scale the fallback year used, which is the predeclared theoretical value). Theory gets this wrong and the fit is what the report uses -- a scale asserted from theory would divide the utilities by too little, and at that value the market-relative candidate loses instead of winning. A run that took the theoretical value on faith would have reported the wrong sign, which is why the first year -- which has no history and falls back to it -- is excluded from the headline and labelled.

CONFIGURATION

Item	Value
Architecture	probit
Paired-diagnostic reference	softmax
Integration	exact_gauss_hermite_quadrature
Shared factors	0
Opening bankroll	10,000,000

PAIRED ARCHITECTURE DIAGNOSTIC

REFERENCE MODEL

softmax

MODEL LOG LOSS

2.030823

REFERENCE LOG LOSS

2.043380

MODEL MINUS
REFERENCE

-0.012557

negative is better

CALIBRATED

-0.007069

both sides given a
temperature

YEARS BETTER

11/11

WORST YEAR

-0.006424

What this comparison holds fixed: the mean utilities. Both links score the identical raw tree utilities, so the only difference is the shock distribution: independent Gumbel against iid normal. Scoring against the Offset model instead would change the mean learner and the link at once. A diagnostic, not an identity. This report's headline is the market delta, because that is the one quantity every published report can be compared on. This delta answers a narrower question and answers it better -- it is the only measurement that separates one change from everything else -- and read as a headline it would mislead, because a model can beat the reference it holds fixed and still lose to the market.

PROMOTION GATES

Gate	Verdict
1. paired delta is negative	pass
2. interval excludes zero favourably	pass
3. no year degrades beyond tolerance	pass
4. survives nested calibration	pass
5. direction stable across seeds	pass
6. improvement exceeds integration error 10x	pass

6 of 6 checkable gate assertions hold. Passing them is not promotion. An interaction diagnostic across predeclared cohorts, reproduction from a clean identified commit, and confirmation on a future prospective period are not derivable from these artifacts. Production remains the five-seed Softmax Offset ensemble until they are met.

PER-SEED DELTA

Seed	Delta
7	-0.013234
17	-0.012556
42	-0.012215
123	-0.012210
256	-0.012571

5 of 5 seeds favour the candidate, and every seed agrees on the sign. Shown individually rather than only pooled, because a mean whose seeds disagree is a different claim from one whose seeds agree.

UTILITY SCALE

setting	value	note
2015	0.779697	fitted on strictly earlier years; fallback to the theoretical value

setting	value	note
2016	1.389269	fitted on strictly earlier years
2017	1.453396	fitted on strictly earlier years
2018	1.439805	fitted on strictly earlier years
2019	1.450206	fitted on strictly earlier years
2020	1.428605	fitted on strictly earlier years
2021	1.426172	fitted on strictly earlier years
2022	1.420903	fitted on strictly earlier years
2023	1.428470	fitted on strictly earlier years
2024	1.426704	fitted on strictly earlier years
2025	1.423754	fitted on strictly earlier years
2026	1.426958	fitted on strictly earlier years

12 row(s) from the sensitivity artifact, in 1 study/studies. Where a study varies an integration setting, a result that moves with it is a property of the integration rather than of the model.

PROVENANCE

Item	Value
Source commit	0422b20c2039
Source tree	dirty at generation
Config digest	unrecorded
Identity source	artifact
Model view	bagged_probit
Artifact root	results/probit_models
Inputs read	8 artifact(s), locked by digest in run_lock.json
Evaluation race keys	7933 races, sha256:342d95ee674808a0d7ae4d0df5b0f958da1898863e25c2898bb71d9f58392e58
Environment: numpy	2.5.2
Environment: pandas	3.0.5
Environment: python	3.12.11
Environment: xgboost	3.4.1

The generating source tree was dirty

This run's provenance records uncommitted changes, so the commit named above does not fully describe the code that produced these numbers. Treat the figures as indicative until the report is regenerated from a clean tree.

Every input this report read is listed in run_lock.json by content digest, together with the lock digest of the run that produced them -- which is what chains a published figure back through the report to the fit and from there to the feature table's own digest.

Limitations and the retrospective-odds warning

Retrospective odds, restated

Every stake, return and bankroll figure in this report is computed against historical result-page odds and official dividends. Those are not an executable pre-race price: they are the prices after the pool closed, and a bet placed at them could not have been placed. Nothing here establishes profitability. A profitability claim requires a locked model shadow-tested against timestamped pre-race odds snapshots.

Restated at the end as well as beside the betting numbers, deliberately. It is the single most load-bearing caveat in this report and the one a reader skimming for a return figure is most likely to miss.

Every test year is also development data

Every year through 2026 has been used as development data in this repository: features were searched, architectures compared and thresholds inspected against these same seasons. The chronological split is real, so a test year is genuinely after its training rows, but no test year is unseen. Treat the forward figures as probability progress rather than as a locked out-of-sample result, and do not select a Kelly level or an edge threshold from this reused history. The confirmation step that this run cannot substitute for is a prospective shadow test against archived pre-race price snapshots.

WHAT ELSE THIS REPORT DOES NOT ESTABLISH

- It measures Probit against the market on the same races. It does not measure this model against the other published models except through that shared yardstick.
- Its intervals condition on the fitted model, so they bound sampling risk and not model risk.
- Every figure is out of sample in time but in sample in *design*: the feature set, the hyperparameters and the staking rules were all chosen with these years visible.
- Nothing here is a shadow test. A profitability claim needs a locked model run forward against timestamped pre-race prices.

Unverified source state

This run recorded a dirty source tree, so its numbers cannot be reproduced from the named commit alone.